

Calculus Practice — Optimization in Neural Networks

8 questions covering forward pass, activation functions, log-loss, backpropagation via the chain rule, and gradient descent weight updates. Then 5 further questions drilling the chain rule through multi-layer networks. Try all of them before checking the answers below.

Practice Questions

Q1 A single neuron has weight $w = 0.5$, bias $b = -1$, and a sigmoid activation $\hat{y} = \sigma(z)$ where $z = wx + b$. For input $x = 2$, compute z and $\hat{y}$.

Q2 The binary cross-entropy (log-loss) for a single training example is:

$$L = -y \log \hat{y} - (1 - y) \log(1 - \hat{y})$$

For a predicted probability $\hat{y} = 0.8$ and true label $y = 1$, compute L (leave your answer in terms of a natural log, or evaluate numerically to 4 decimal places).

Q3 Show, step by step using the chain rule, that:

$$\frac{\partial L}{\partial z} = \hat{y} - y$$

where L is the log-loss, $\hat{y} = \sigma(z)$, and y is the true label. (You may use the result $\sigma'(z) = \hat{y}(1 - \hat{y})$ from Week 1.)

Q4 Given the result from Q3, find $\frac{\partial L}{\partial w}$ and $\frac{\partial L}{\partial b}$ for the neuron $z = wx + b$. Then evaluate them numerically for $\hat{y} = 0.7$, $y = 1$, and $x = 3$.

Q5 Using your answers from Q4, perform one gradient descent step with learning rate $\alpha = 0.1$, starting from $w = 0.5$, $b = -1$.

Q6 The ReLU activation function is defined as:

$$\text{ReLU}(z) = \max(0, z)$$

Write down its derivative $\text{ReLU}'(z)$, and explain in one or two sentences why ReLU is often preferred over sigmoid in deep networks.

Q7 The tanh activation function has derivative:

$$\frac{d}{dz} \tanh(z) = 1 - \tanh^2(z)$$

(a) Verify this is consistent with the identity $\tanh(z) = 2\sigma(2z) - 1$ by differentiating both sides. (b) Compare the output ranges of $\sigma(z)$ and $\tanh(z)$, and explain one scenario where tanh is preferred.

Q8 For a linear output neuron ($\hat{y} = wx + b$, no activation) with squared-error loss $L = (\hat{y} - y)^2$, derive $\frac{\partial L}{\partial w}$ and $\frac{\partial L}{\partial b}$ using the chain rule. Then evaluate them for $w = 1$, $b = 0$, $x = 3$, $y = 2$. (Compare to the regression gradient you derived in earlier notebooks — it should look familiar.)

Part B: Chain Rule Through Multiple Layers

Q9 Consider a two-layer network with a single input x :

$$z^{[1]} = w^{[1]}x + b^{[1]}, \quad a^{[1]} = \sigma(z^{[1]}), \quad z^{[2]} = w^{[2]}a^{[1]} + b^{[2]}, \quad \hat{y} = \sigma(z^{[2]})$$

with log-loss L and true label $y = 1$. Without substituting any numbers, write out the full chain rule expression for $\frac{\partial L}{\partial w^{[1]}}$ as a product of four terms, naming each one.

Q10 Using the same two-layer network as Q9, compute the **full forward and backward pass numerically** for:

$$x = 1, \quad w^{[1]} = 2, \quad b^{[1]} = 0, \quad w^{[2]} = 1, \quad b^{[2]} = 0, \quad y = 1$$

Find $z^{[1]}$, $a^{[1]}$, $z^{[2]}$, $\hat{y}$, L , and then $\frac{\partial L}{\partial w^{[2]}}$ and $\frac{\partial L}{\partial w^{[1]}}$. (Round intermediate values to 4 decimal places.)

Q11 Extend the chain to **three layers**: $a^{[1]} = \sigma(w^{[1]}x)$, $a^{[2]} = \sigma(w^{[2]}a^{[1]})$, $a^{[3]} = \sigma(w^{[3]}a^{[2]})$, $L = (a^{[3]} - y)^2$. Write out the full chain rule expression for $\frac{\partial L}{\partial w^{[1]}}$ as a product of six terms.

Q12 The sigmoid derivative satisfies $\sigma'(z) = \sigma(z)(1 - \sigma(z)) \leq \frac{1}{4}$ for all z . Using the three-layer chain from Q11, show that:

$$\left| \frac{\partial L}{\partial w^{[1]}} \right| \leq \frac{1}{2} \cdot \left(\frac{1}{4} \right)^3 \cdot |w^{[3]}| \cdot |w^{[2]}| \cdot |x|$$

and explain what happens to this bound as the number of layers grows. (This is the *vanishing gradient problem*.)

Q13 Replace the hidden-layer sigmoid in the two-layer network of Q10 with a ReLU: $a^{[1]} = \text{ReLU}(z^{[1]})$, keeping the output as $\hat{y} = \sigma(z^{[2]})$ and the loss as log-loss. Using the **same weights and input as Q10**, compute $\frac{\partial L}{\partial w^{[1]}}$ and compare your answer to Q10 — why is it larger in magnitude?

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Answers & Solutions

A1. $z = wx + b = 0.5(2) + (-1) = 0$.

$$\hat{y} = \sigma(0) = \frac{1}{1 + e^0} = 0.5$$

A z -value of 0 always maps to exactly 0.5 with the sigmoid — the neuron is maximally uncertain.

A2. With $y = 1$ the second term vanishes:

$$L = -\log(0.8) \approx 0.2231$$

A3. Apply the chain rule in two steps.

Step 1 — differentiate L with respect to $\hat{y}$:

$$\frac{\partial L}{\partial \hat{y}} = -\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}}$$

Step 2 — multiply by $\frac{\partial \hat{y}}{\partial z} = \hat{y}(1-\hat{y})$:

$$\frac{\partial L}{\partial z} = \left(-\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}}\right) \hat{y}(1-\hat{y}) = -y(1-\hat{y}) + (1-y)\hat{y} = \hat{y} - y$$

A4. Since $z = wx + b$, the remaining partials are $\frac{\partial z}{\partial w} = x$ and $\frac{\partial z}{\partial b} = 1$. By the chain rule:

$$\frac{\partial L}{\partial w} = (\hat{y} - y) \cdot x, \quad \frac{\partial L}{\partial b} = (\hat{y} - y)$$

With $\hat{y} = 0.7$, $y = 1$, $x = 3$:

$$\frac{\partial L}{\partial w} = (-0.3)(3) = -0.9, \quad \frac{\partial L}{\partial b} = -0.3$$

A5.

$$w \leftarrow 0.5 - 0.1(-0.9) = 0.59, \quad b \leftarrow -1.0 - 0.1(-0.3) = -0.97$$

A6.

$$\text{ReLU}'(z) = \begin{cases} 1 & z > 0 \\ 0 & z < 0 \end{cases}$$

ReLU is preferred in deep networks because its derivative is exactly 1 for all positive inputs, so gradients don't shrink as they travel backwards through many layers. With sigmoid or tanh, the derivative is always $\leq \frac{1}{4}$ (sigmoid) or ≤ 1 (tanh), so multiplying many such

derivatives together causes them to vanish — the *vanishing gradient problem*. ReLU avoids it for the active ($z > 0$) units.

A7. (a) Differentiating $\tanh(z) = 2\sigma(2z) - 1$ using the chain rule:

$$\frac{d}{dz} \tanh(z) = 2 \cdot \sigma'(2z) \cdot 2 = 4\sigma(2z)(1 - \sigma(2z))$$

Since $\sigma(2z) = \frac{1+\tanh(z)}{2}$ and $1 - \sigma(2z) = \frac{1-\tanh(z)}{2}$:

$$4 \cdot \frac{1 + \tanh(z)}{2} \cdot \frac{1 - \tanh(z)}{2} = 1 - \tanh^2(z) \quad \checkmark$$

(b) $\sigma(z) \in (0, 1)$ while $\tanh(z) \in (-1, 1)$. Because $\tanh$ is zero-centred, the gradients it passes to earlier layers have a balanced sign distribution, which can make learning faster. In practice, $\tanh$ is often preferred in hidden layers when zero-centred activations are beneficial, while σ is reserved for binary classifier output layers.

A8. By the chain rule with $\hat{y} = wx + b$:

$$\frac{\partial L}{\partial w} = 2(\hat{y} - y) \cdot x, \quad \frac{\partial L}{\partial b} = 2(\hat{y} - y)$$

With $w = 1$, $b = 0$, $x = 3$, $y = 2$: $\hat{y} = 3$, $\hat{y} - y = 1$, so $\frac{\partial L}{\partial w} = 6$ and $\frac{\partial L}{\partial b} = 2$. This is exactly the regression gradient from earlier notebooks.

Part B Answers

A9. Expanding the chain rule one link at a time:

$$\begin{aligned} \frac{\partial L}{\partial w^{[1]}} &= \underbrace{\frac{\partial L}{\partial z^{[2]}}}_{\hat{y}-y} \cdot \underbrace{\frac{\partial z^{[2]}}{\partial a^{[1]}}}_{w^{[2]}} \cdot \underbrace{\frac{\partial a^{[1]}}{\partial z^{[1]}}}_{a^{[1]}(1-a^{[1]})} \cdot \underbrace{\frac{\partial z^{[1]}}{\partial w^{[1]}}}_x \\ &= (\hat{y} - y) \cdot w^{[2]} \cdot a^{[1]}(1 - a^{[1]}) \cdot x \end{aligned}$$

The gradient must travel all the way back through the output activation, the second weight, and the hidden activation before finally reaching $w^{[1]}$.

A10. *Forward pass:*

$$z^{[1]} = 2(1) + 0 = 2, \quad a^{[1]} = \sigma(2) \approx 0.8808$$

$$z^{[2]} = 1(0.8808) + 0 = 0.8808, \quad \hat{y} = \sigma(0.8808) \approx 0.7070$$

$$L = -\log(0.7070) \approx 0.3467$$

Backward pass (layer 2):

$$\frac{\partial L}{\partial z^{[2]}} = \hat{y} - y = 0.7070 - 1 = -0.2930$$

$$\frac{\partial L}{\partial w^{[2]}} = (-0.2930)(0.8808) \approx -0.2581$$

Backward pass (layer 1):

$$\frac{\partial L}{\partial a^{[1]}} = (-0.2930)(1) = -0.2930$$

$$\frac{\partial a^{[1]}}{\partial z^{[1]}} = a^{[1]}(1 - a^{[1]}) = (0.8808)(0.1192) \approx 0.1050$$

$$\frac{\partial L}{\partial w^{[1]}} = (-0.2930)(0.1050)(1) \approx -0.0308$$

Notice $\left| \frac{\partial L}{\partial w^{[1]}} \right| \approx 0.031$ is much smaller than $\left| \frac{\partial L}{\partial w^{[2]}} \right| \approx 0.258$ — the gradient has already shrunk considerably after passing through just one sigmoid.

A11. Expanding the chain fully:

$$\begin{aligned} \frac{\partial L}{\partial w^{[1]}} &= \underbrace{\frac{\partial L}{\partial a^{[3]}}}_{2(a^{[3]}-y)} \cdot \underbrace{\frac{\partial a^{[3]}}{\partial z^{[3]}}}_{\sigma'(z^{[3]})} \cdot \underbrace{\frac{\partial z^{[3]}}{\partial a^{[2]}}}_{w^{[3]}} \cdot \underbrace{\frac{\partial a^{[2]}}{\partial z^{[2]}}}_{\sigma'(z^{[2]})} \cdot \underbrace{\frac{\partial z^{[2]}}{\partial a^{[1]}}}_{w^{[2]}} \cdot \underbrace{\frac{\partial a^{[1]}}{\partial z^{[1]}}}_{\sigma'(z^{[1]})} \cdot \underbrace{\frac{\partial z^{[1]}}{\partial w^{[1]}}}_x \\ &= 2(a^{[3]} - y) \cdot \sigma'(z^{[3]}) \cdot w^{[3]} \cdot \sigma'(z^{[2]}) \cdot w^{[2]} \cdot \sigma'(z^{[1]}) \cdot x \end{aligned}$$

A12. From the A11 expression, taking absolute values and using $\sigma'(z) \leq \frac{1}{4}$ and $|a^{[3]} - y| \leq 1$:

$$\left| \frac{\partial L}{\partial w^{[1]}} \right| \leq 2(1) \cdot \frac{1}{4} \cdot |w^{[3]}| \cdot \frac{1}{4} \cdot |w^{[2]}| \cdot \frac{1}{4} \cdot |x| = \frac{1}{2} \cdot \left(\frac{1}{4} \right)^3 \cdot |w^{[3]}| \cdot |w^{[2]}| \cdot |x|$$

For a network with n sigmoid layers the bound contains $\left(\frac{1}{4}\right)^n$. Since $\frac{1}{4} < 1$ this decays **exponentially** with depth: with 10 sigmoid layers the factor is at most $(1/4)^{10} \approx 10^{-6}$, meaning the gradient reaching the first layer is essentially zero. This is why deep networks with sigmoid activations barely learn in early layers — the vanishing gradient problem.

A13. Forward pass (ReLU hidden layer):

$$z^{[1]} = 2, \quad a^{[1]} = \text{ReLU}(2) = 2$$

$$z^{[2]} = 1(2) + 0 = 2, \quad \hat{y} = \sigma(2) \approx 0.8808, \quad L \approx 0.1269$$

Backward pass:

$$\frac{\partial L}{\partial z^{[2]}} = 0.8808 - 1 = -0.1192$$

$$\frac{\partial L}{\partial a^{[1]}} = (-0.1192)(1) = -0.1192$$

$$\frac{\partial a^{[1]}}{\partial z^{[1]}} = \text{ReLU}'(2) = 1$$

$$\frac{\partial L}{\partial w^{[1]}} = (-0.1192)(1)(1) = -0.1192$$

Compared to Q10's -0.0308 , the gradient is about $4\times$ larger in magnitude. The reason: in Q10 the sigmoid hidden layer contributed $\sigma'(z^{[1]}) \approx 0.105$ as a multiplicative factor; here ReLU contributes exactly 1, so no information is lost when passing the gradient back through that layer.